

Instructions.

1. **Due May 18, 11:59 PM uploaded to Blackboard as a single pdf and a single ZIP archive. This deadline is firm. Late uploads will not be accepted.**
2. This project will be done in a team of two members. You may **NOT** discuss or consult with any students outside your team.
3. All work must be typeset in L^AT_EX. Handwritten or scanned submissions will not be accepted.
4. All figures must have axis labels and descriptive captions. Tables, if any, must have descriptive captions.
5. This document may be updated if a mistake is found or further clarification is needed. A Blackboard announcement will be posted as soon as this document is updated.
6. Submission guidelines.
 - (a) Label your report as ENME408_S26_Project04_<lastnames>.pdf.
 - (b) Include all figures and tables with captions in your report.
 - (c) You may include MATLAB scripts and representative outputs as Appendices. Label the appendices and cite them in your report appropriately.
 - (d) Place all TEX files, MATLAB scripts, and Simulink models in a folder, then archive it. Label your archive as ENME408_S26_Project04_<lastnames>.zip.
 - (e) Upload your pdf file and the archive folder to Blackboard before the deadline.
7. Honor code: “I have neither received nor given aid to anyone outside my team during this project.”

Sign/print your names here.

If you cannot sign/print in the box above, please write the honor code on the first page of your submission and sign it.

8. Reports with substantial similarity will be reported to the UMBC Academic Conduct Committee.
9. Include a **credit author statement** in your report. The statement recognizes individual author contributions and facilitates collaboration. For example,
Author 1: Methodology, Software, Visualization
Author 2: Data curation, Writing, Original draft preparation, Validation.

Kalman Filter for 2D Position and Velocity Estimation

Objective. The objective of this project is to implement a discrete-time Kalman filter to estimate the state of a two-dimensional moving object from noisy position and acceleration measurements. The project is designed to illustrate the prediction–correction structure of the Kalman filter in a setting where the true trajectory is known and can be used for validation.

Problem Description

Consider a robot modeled as a point mass moving in a two-dimensional plane with position

$$q(t) = \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} \in \mathbb{R}^2. \quad (1)$$

The robot acceleration in the inertial frame and the robot position are measured at discrete sampling instants

$$t_k = k\Delta t, \quad (2)$$

at a sampling frequency of 100 Hz, that is, $\Delta t = 0.01$ s. The measured acceleration is

$$a_{m,k} = \begin{bmatrix} a_{1,m,k} \\ a_{2,m,k} \end{bmatrix}, \quad (3)$$

and the measured position is

$$q_{m,k} = \begin{bmatrix} q_{1,m,k} \\ q_{2,m,k} \end{bmatrix}. \quad (4)$$

Both measurements are corrupted by additive Gaussian noise and are provided in the file

[circular_motion_noisy_position_acceleration.csv](#)

The provided CSV file has the following structure.

Column	Description
Time_s	Time stamp t_k [s]
q1_true	True position $q_{1,k}$
q2_true	True position $q_{2,k}$
v1_true	True velocity $\dot{q}_{1,k}$
v2_true	True velocity $\dot{q}_{2,k}$
a1_true	True acceleration $a_{1,k}$
a2_true	True acceleration $a_{2,k}$
q1_meas	Noisy position measurement $q_{1,m,k}$
q2_meas	Noisy position measurement $q_{2,m,k}$
a1_meas	Noisy acceleration measurement $a_{1,m,k}$
a2_meas	Noisy acceleration measurement $a_{2,m,k}$

The columns labeled true are provided only for validation and comparison. They should not be used inside the Kalman filter implementation.

Your objective is to use these noisy acceleration and position measurements to estimate the full state

$$x_k \triangleq \begin{bmatrix} q_{1,k} \\ q_{2,k} \\ \dot{q}_{1,k} \\ \dot{q}_{2,k} \end{bmatrix}. \quad (5)$$

Project Tasks

1. **Measurement visualization.** Load the provided CSV file containing the noisy position and acceleration measurements.

- (a) Plot the noisy position measurements in the (q_1, q_2) plane.
- (b) Plot the measured positions and accelerations versus time.
- (c) Briefly comment on the quality of the measurements and the effect of measurement noise.

2. **Discrete-time dynamics model.** The translational dynamics are governed by

$$\ddot{q}(t) = a(t). \quad (6)$$

Define the state and input as

$$x(t) = \begin{bmatrix} q_1(t) \\ q_2(t) \\ \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix}, \quad u(t) = \begin{bmatrix} a_1(t) \\ a_2(t) \end{bmatrix}. \quad (7)$$

- (a) Derive the continuous-time state-space model

$$\dot{x} = A_c x + B_c u. \quad (8)$$

- (b) Use MATLAB `c2d()` with zero-order hold discretization and sampling time $\Delta t = 0.01$ s to determine the discrete-time model

$$x_{k+1} = A x_k + B u_k. \quad (9)$$

3. **Open-loop propagation using noisy acceleration.** Using the noisy acceleration measurements from the CSV file as the system input, propagate the discrete-time model forward in time without measurement correction.

- (a) Use the noisy acceleration measurements as input

$$u_k = a_{m,k}. \quad (10)$$

- (b) Use the initial measured position and zero initial velocity.
- (c) Plot the resulting trajectory and compare it with the noisy position measurements.
- (d) Explain why double integration of noisy acceleration causes drift in the position estimate.

4. **Kalman filter design.** Assume the available position measurements are

$$y_k = \begin{bmatrix} q_{1,m,k} \\ q_{2,m,k} \end{bmatrix} = C x_k + v_k. \quad (11)$$

- (a) Determine the output matrix C .
- (b) Select reasonable process noise covariance Q and measurement noise covariance R . Justify your choice. Relate your choice of R to the measurement noise standard deviation observed in the data.
- (c) Implement a discrete-time Kalman filter for state estimation.
- (d) Clearly show the prediction step

$$\hat{x}_{k+1|k} = A \hat{x}_{k|k} + B u_k \quad (12)$$

$$P_{k+1|k} = A P_{k|k} A^T + Q \quad (13)$$

and the measurement update step

$$K_{k+1} = P_{k+1|k} C^T (C P_{k+1|k} C^T + R)^{-1} \quad (14)$$

$$r_{k+1} = y_{k+1} - C \hat{x}_{k+1|k} \quad (15)$$

$$\hat{x}_{k+1|k+1} = \hat{x}_{k+1|k} + K_{k+1} r_{k+1} \quad (16)$$

$$P_{k+1|k+1} = (I - K_{k+1} C) P_{k+1|k} \quad (17)$$

5. Results and discussion.

(a) Plot and compare the following trajectories:

- true trajectory
- noisy position measurements
- open-loop propagation using noisy acceleration
- Kalman filter estimate

(b) Plot

$$q_1(t), \quad q_2(t), \quad \dot{q}_1(t), \quad \dot{q}_2(t) \quad (18)$$

versus time.

- (c) Discuss how the Kalman filter improves the estimate compared to direct acceleration integration.
- (d) Discuss the effect of changing the covariance matrices Q and R .